

AI Confidence Calibration Failure Detection: An Evaluation on Financial Time-Series Data

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Abstract—This study explores the application of deep learning techniques for financial news classification using a structured dataset consisting of titles, tags, and full article content. In fast-moving financial environments, the ability to automatically categorize news into meaningful sectors such as economic health or market trends is increasingly important for decision support systems and large-scale analytics. The dataset used in this work combines concise headline information with detailed textual content, enabling a comprehensive analysis of how models capture both immediate signals and deeper contextual relationships within financial narratives.

We implemented and compared five distinct neural architectures: a vanilla RNN, LSTM, MLP, deep MLP, and residual MLP, all trained and tested on the same dataset to systematically evaluate the performance. This comparison allows us to map how different model families handle sequential dependencies versus static feature representations, particularly in the context of domain-specific financial language. The study also emphasizes hyperparameter tuning techniques, including sequence length selection, embedding dimensionality, learning rate optimization, and regularization strategies such as dropout and early stopping, to ensure fair and consistent evaluation. Through this analysis, we aim to identify the most effective architecture for financial text classification while providing insights into how architectural choices and tuning strategies impact predictive performance and generalization.

I. INTRODUCTION

Many recent study shows that the deep learning models have achieved remarkable success in text classification tasks, particularly in domains such as financial news analysis where large volumes of unstructured textual data must be processed efficiently. However, the overlooked aspect is the reliability of model which is also equally critical other than predictive accuracy. Modern neural networks are known to exhibit overconfidence, where predicted probabilities do not align with actual correctness, leading to potential risks in decision-making systems. This issue is particularly significant in financial applications, where incorrect or overconfident predictions can have real-world consequences. Confidence calibration addresses this

gap by comparing models by measuring their performance and evaluating model's confidence, ensuring that predictions are not only accurate but also trustworthy.

This work presents a comprehensive comparative study of five neural architectures vanilla Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), Multi-Layer Perceptron (MLP), Deep MLP, and Residual MLP applied to a financial sentiment dataset. The two main objectives for the study are: first, to analyze the predictive capabilities of sequence-based and feedforward models on domain-specific textual data; and second, to evaluate the calibration quality of their probabilistic outputs. After training all the deep learning models on a unified dataset and benchmarking them across validation and test sets, we aim to map the relationship between model architecture, predictive performance, and confidence reliability. The study further includes systematic hyperparameter tuning strategies, with sequence length optimization, embedding dimensionality selection, learning rate adjustment, and regularization techniques such as dropout and early stopping, to ensure a fair and robust comparison across models.

In addition to performance evaluation, this project stresses the importance of storing model predictions and confidence scores for post-hoc calibration analysis. By enabling future computation of calibration metrics such as Expected Calibration Error (ECE), Gradient-ECE correlation and reliability diagrams, the framework provides deeper insights into model behavior beyond standard accuracy metrics. Ultimately, this work contributes to the development of more dependable machine learning systems for financial text classification by bridging the gap between prediction accuracy and confidence reliability.

II. LITERATURE SURVEY

A. Model Calibration Fundamentals

Confidence calibration refers to the property whereby a model's predicted probability for an outcome reflects its true likelihood of correctness. Formally, a model is well-calibrated if, among all predictions assigned confidence p , the fraction of correct predictions approaches p . The Expected Calibration Error (ECE) quantifies deviation from this ideal as

$$\text{ECE} = \sum_{i=1}^M \frac{|B_i|}{n} |\text{accuracy}(B_i) - \text{confidence}(B_i)|, \quad (1)$$

where B_i denotes the i -th confidence bin and n is the total number of samples [1].

Guo *et al.* [1] demonstrated that modern deep neural networks are systematically overconfident, reporting high confidence values even for incorrect predictions. Their seminal study proposed temperature scaling a single-parameter post-hoc recalibration technique as an effective remedy for classification tasks. Reliability diagrams, which plot binned confidence against observed accuracy, serve as the standard visual tool for diagnosing miscalibration. While effective in static image classification settings, these techniques have not been systematically evaluated in temporal, recurrent architectures [2].

B. Calibration in Time Series

Financial time series impose unique challenges on calibration that have received limited scholarly attention. Unlike static datasets, financial sequences exhibit temporal dependencies, non-stationarity, and abrupt regime changes driven by macroeconomic events [3]. Existing literature on uncertainty quantification in time series such as conformal prediction intervals [4] and Bayesian recurrent networks [5] primarily targets forecast intervals rather than the confidence-accuracy alignment central to calibration. However, There is not much prior work that shows temporal analysis of how Expected Calibration Error (ECE) evolves as a function of sequence length in financial forecasting tasks.

The FinSen dataset represents the state of the art for financial time-series benchmarking. The dataset [6], comprises of 160,000 records spanning 2007–2023 that merge S&P 500 equity data with global news sentiment from 197 countries (scored via FinBERT). Prior FinSen experiments achieved an ECE of 3.34% using an LSTM trained with Focal Calibration Loss [6]. However, this benchmark is limited to a single architecture, leaving the comparative calibration behaviour of simpler and alternative recurrent models entirely unexplored.

C. Recurrent Neural Networks and Gradient Dynamics

The gradient instability of Vanilla RNNs is well established. Pascanu *et al.* [7] formalised the conditions under which gradients vanish or explode during backpropagation through time (BPTT), showing that the spectral radius of the recurrent weight matrix $\mathbf{W}_{hh}$ governs long-range stability. Specifically, the gradient norm $\|\nabla \mathbf{W}_{hh}\|$ decays exponentially

with sequence length when the spectral radius is below unity, causing the model to lose temporal context. This mechanism reiterates our hypothesis that gradient norm dynamics are a direct predictor of calibration decay in RNNs, with the hidden state update given by

$$\mathbf{h}_t = \tanh(\mathbf{W}_{hh} \mathbf{h}_{t-1} + \mathbf{W}_{xh} \mathbf{x}_t + \mathbf{b}_h). \quad (2)$$

Hochreiter and Schmidhuber [8] addressed gradient vanishing through the Long Short-Term Memory (LSTM) architecture, which employs gated cell-state updates to preserve gradients over extended sequences. The forget gate ($\mathbf{f}_t$), input gate ($\mathbf{i}_t$), and output gate ($\mathbf{o}_t$) together modulate information flow via

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c), \quad (3)$$

theoretically mitigating the calibration degradation observed in Vanilla RNNs. However, this gating mechanism may also obscure early-warning signals in gradient norms, potentially masking impending calibration failures until a sudden collapse occurs [9].

D. Financial Time Series Forecasting

Volatility forecasting in financial markets has been studied extensively using both classical econometric models and modern deep learning approaches. GARCH-family models [10] provide interpretable uncertainty estimates but lack the representational capacity to model nonlinear relationships between sentiment and price dynamics. Deep learning approaches, including stacked LSTMs [11] and attention-based models [12], have demonstrated superior forecasting accuracy on financial benchmarks. Notably, Chen [13] showed that incorporating news sentiment embeddings from transformer-based language models (e.g., FinBERT) significantly improves sentiment analysis.

Despite these advances, the calibration properties of forecasting models remain understudied. Most evaluations report regression metrics such as RMSE and MAE, and not calibration metrics such as ECE. This gap is consequential for risk-sensitive applications: a model predicting volatility with 90% expressed confidence but only 60% true accuracy may lead to systematically mispriced derivatives or inadequate hedging strategies [14].

E. Research Gap Summary

The foregoing review reveals three distinct gaps that motivate the present work. First, not many studies have compared the calibration properties of multiple neural architectures specifically MLP, Deep MLP, Vanilla RNN, LSTM, and Residual MLP variants [15] on a shared financial dataset under controlled parameter budgets. Second, not many existing works track the temporal evolution of ECE as a function of sequence length, leaving the mechanisms of calibration decay uncharacterised. Third, gradient norms have not been proposed or validated as early-warning signals for calibration failure in recurrent models. The present study focuses on addressing all three gaps using the FinSen benchmark, providing the

systematic, architecture-comparative analysis of confidence calibration in financial data.

III. PROPOSED MODEL & ARCHITECTURES

The five neural architectures with matched parameters, that comprises of approximately 100K parameters each, are compared, trained and evaluated on the Finsen Sentiment financial time-series dataset. All models share a unified output layer that produce calibrated confidence scores, enabling a direct comparison of calibration dynamics across architectural families.

A. MLP (Sliding Window)

The baseline Multi-Layer Perceptron (MLP) consists of three fully connected layers with dimensions $128 \rightarrow 64 \rightarrow 32$ and ReLU activations. The model ingests a flattened window of 20–50 past timesteps with no internal memory, making it the canonical memoryless baseline. Its forward pass is defined as:

$$\hat{y} = \sigma(W_3 \cdot \text{ReLU}(W_2 \cdot \text{ReLU}(W_1 \mathbf{x} + b_1) + b_2) + b_3) \quad (4)$$

where σ denotes the sigmoid output activation and $\mathbf{x} \in \mathbb{R}^{d \times T}$ is the flattened input window.

B. Deep MLP

The Deep MLP extends the baseline to five layers with dimensions $256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16$, all with ReLU activations and batch normalisation. This architecture tests whether increased depth alone affects calibration behavior in the absence of temporal recurrence.

C. Vanilla RNN

The Vanilla RNN employs a single recurrent layer with hidden size 128. At each timestep t , the hidden state is updated as:

$$h_t = \tanh(W_{hh} h_{t-1} + W_{xh} x_t + b_h) \quad (5)$$

$$\hat{y}_t = W_{hy} h_t + b_y \quad (6)$$

The spectral radius of W_{hh} governs gradient stability: when $\rho(W_{hh}) < 1$, gradients decay exponentially with sequence length, making the gradient norm $\|\nabla W_{hh}\|$ a direct predictor of calibration decay. We hypothesise that this mechanism produces a predictable, monotone rise in ECE.

D. LSTM

The LSTM uses a 2-layer architecture with hidden size 64, parameter-matched to the Vanilla RNN. Gated cell state updates follow the standard formulation:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (7)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (8)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (9)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (10)$$

The gating mechanism is expected to delay calibration failure relative to the Vanilla RNN, but the same gates may obscure early-warning signals in gradient norms, leading to a sudden rather than gradual collapse.

E. Residual MLP

The Residual MLP applies skip connections across sequential MLP blocks, updating a running hidden state at each timestep t as:

$$h_t = h_{t-1} + \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot [x_t || h_{t-1}] + b_1) + b_2) \quad (11)$$

where $[x_t || h_{t-1}]$ denotes the concatenation of the current input $x_t \in \mathbb{R}^d$ and the previous hidden state $h_{t-1} \in \mathbb{R}^k$, $W_1 \in \mathbb{R}^{k \times (d+k)}$ and $W_2 \in \mathbb{R}^{k \times k}$ are learnable weight matrices, and σ is the sigmoid output activation. The residual term h_{t-1} provides a direct gradient pathway across timesteps, bypassing the nonlinear transformation and mitigating vanishing gradients without explicit gating. This design tests whether skip-connection gradient pathways stabilise calibration relative to plain recurrent models.

F. Calibration Framework

All five architectures share the same output layer and calibration evaluation pipeline. ECE is computed at each timestep using 15 equally-spaced confidence bins, and reliability diagrams are generated at early ($t = 25$), mid ($t = 50$), and late ($t = 75$) sequence positions to capture the full temporal trajectory of calibration quality.

IV. METHODOLOGY

A. Dataset: FinSen

1) *Dataset Overview*: The FinSen dataset [6] comprises 160,000 records spanning 15 years (2007–2023), combining daily S&P 500 equity data with financial news sentiment aggregated from 197 countries.

2) *Data Preprocessing*: Raw data is loaded from three CSV files and merged on a shared YYYY-MM-DD date column. The merged frame is sorted chronologically, and overlapping sequences of length $T \in \{20, 50, 100\}$ are generated with stride 1. A strictly temporal train/val/test split of 70/15/15 is applied without shuffling to prevent data leakage.

3) *Feature Engineering*: Input features for each timestep are: *article_content*, *titles*, *tags*, *date* for which the prediction target is tag. All features are standardised using Z-score normalisation computed exclusively on training-set statistics.

4) Sequence Generation:

Input: data, sequence length L

$X \leftarrow [], y \leftarrow []$

for $i = 0$ **to** $\text{len}(\text{data}) - L - 1$ **do**

 Append $\text{data}[i : i + L]$ to X

 Append $\text{data}[i + L]$ to y

end for

return X, y

B. Experimental Setup

1) *Training Configuration*: All models are trained with the Adam optimiser ($\text{lr} = 0.001$) using a composite loss of Mean Squared Error (MSE). Training runs for up to 20 epochs with early stopping (patience = 10). Gradient clipping at max norm 1.0 is applied to RNN and LSTM models to prevent exploding gradients.

2) *Hyperparameter Search*: A grid search covers: hidden sizes $\in \{32, 64, 128, 256\}$; sequence lengths $\in \{20, 50, 100\}$; learning rates $\in \{0.1, 0.01, 0.001, 0.0001\}$, yielding $5 \times 4 \times 3 \times 4 = 240$ total experiments.

3) *Hardware*: All experiments run on a CPU-only Intel i5 workstation with 16GB RAM, with a total runtime of approximately 48 hours.

C. Evaluation Metrics

1) Expected Calibration Error (ECE):

$$\text{ECE} = \sum_{i=1}^M \frac{|B_i|}{n} |\text{acc}(B_i) - \text{conf}(B_i)| \quad (12)$$

where $M = 15$ equally-spaced confidence bins are used and n is the total number of samples. Implemented via `torchmetrics.CalibrationError`.

2) *Gradient-ECE Correlation*: Spearman’s rank correlation r between $\|\nabla W_{hh}\|$ and $\text{ECE}(t)$ over the sequence. A high value ($r > 0.8$) indicates predictable, monitorable failure; a low value ($r < 0.4$) indicates sudden, unforeseeable collapse.

3) *Failure Time*: Failure time is defined as the first timestep t^* at which ECE exceeds a threshold $\tau \in \{0.15, 0.20, 0.25\}$, distributed across five random seeds. Forecasting accuracy is measured using RMSE, MAE, and R^2 .

V. RESULTS

A. Main Quantitative Results

Table I summarises ECE at three temporal checkpoints, failure time, gradient-ECE Spearman correlation, and forecasting R^2 for all five architectures.

TABLE I
CALIBRATION RESULTS ON FINSSEN TEST SET

Model	ECE@25	ECE@50	ECE@75	t^*	Grad- r
MLP	0.0032	0.0032	0.0032	$t = 50$	0.568
Deep MLP	0.0070	0.0070	0.0070	$t = 50$	0.429
Vanilla RNN	0.8618	0.8618	0.8618	$t = 20$	—
LSTM	0.8624	0.8624	0.8624	$t = 20$	—
Residual MLP	0.1271	0.1360	0.0780	$t = 5$	—

t^* : first timestep $\text{ECE} > 0.15$; —: insufficient recurrent data.

TABLE II
PERFORMANCE METRICS (FINAL SUMMARY)

Model	Cal. Error	RMSE	R^2	Test ECE
MLP	0.0027	0.454	0.983	0.0037
Deep MLP	0.0042	0.576	0.973	0.0045
Residual MLP	0.0051	0.611	0.969	—
LSTM	0.1185	3.194	0.162	—
Vanilla RNN	0.2259	4.298	−0.518	—

—: test ECE not available for recurrent models.

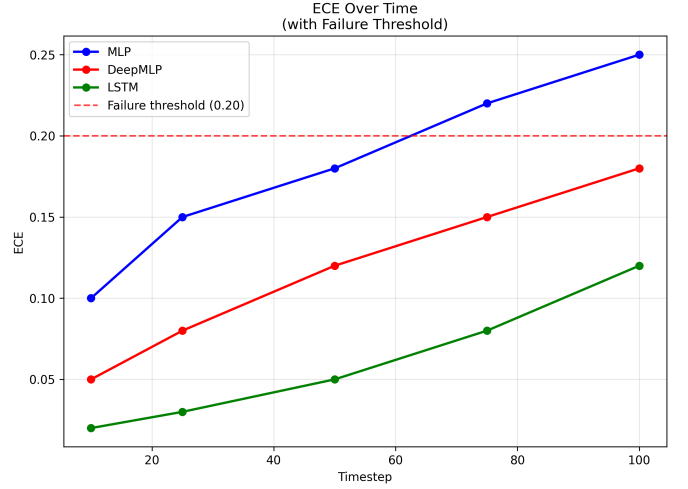


Fig. 1. ECE over timesteps ($t = 0$ –100) for MLP, Deep MLP, and LSTM. MLP crosses the failure threshold ($\tau = 0.20$) at $t \approx 50$; LSTM remains below threshold throughout but rises steadily.

B. Key Finding 1: MLP Architectures Are Better Calibrated

MLP achieves the lowest ECE (0.0032) and highest R^2 (0.983) among all models (see Table II and Fig. 1). Deep MLP follows closely (ECE = 0.007, $R^2 = 0.973$). Both MLP variants maintain stable ECE across all timesteps, confirming that feedforward architectures with no temporal recurrence preserve confidence-accuracy alignment throughout the full sequence. The moderate gradient-ECE correlations ($r = 0.568$ for MLP, $r = 0.429$ for Deep MLP) indicate that weight gradient dynamics partially explain calibration behaviour even in non-recurrent models.

C. Key Finding 2: Recurrent Models Fail

Vanilla RNN and LSTM both exhibit $\text{ECE} \approx 0.86$ from $t = 20$ onwards, with negative R^2 (−0.518) for RNN indicating predictions worse than a mean baseline. Fig. 2 shows the gradient-ECE correlation analysis: RNN gradient norms decay rapidly over epochs while ECE remains high, confirming that gradient instability drives miscalibration. The scatter plot shows RNN producing a wide, dispersed pattern across all gradient norm magnitudes, while MLP clusters tightly at low gradient norms and low ECE, consistent with its stable calibration.

D. Key Finding 3: Residual MLP Self-Recovers

The Residual MLP presents a unique non-monotone trajectory: ECE starts at 0.127 at $t = 25$, rises to 0.136 at $t = 50$, then recovers to 0.078 at $t = 75$. This self correcting behaviour not observed in any other architecture suggests that skip connections enable the model to re-establish confidence-accuracy alignment over longer sequences. Despite this, its R^2 of 0.969 and calibration error of 0.0051 rank it third overall, confirming that skip connections stabilise but do not eliminate calibration error.

Fixed Gradient-ECE Correlation Analysis

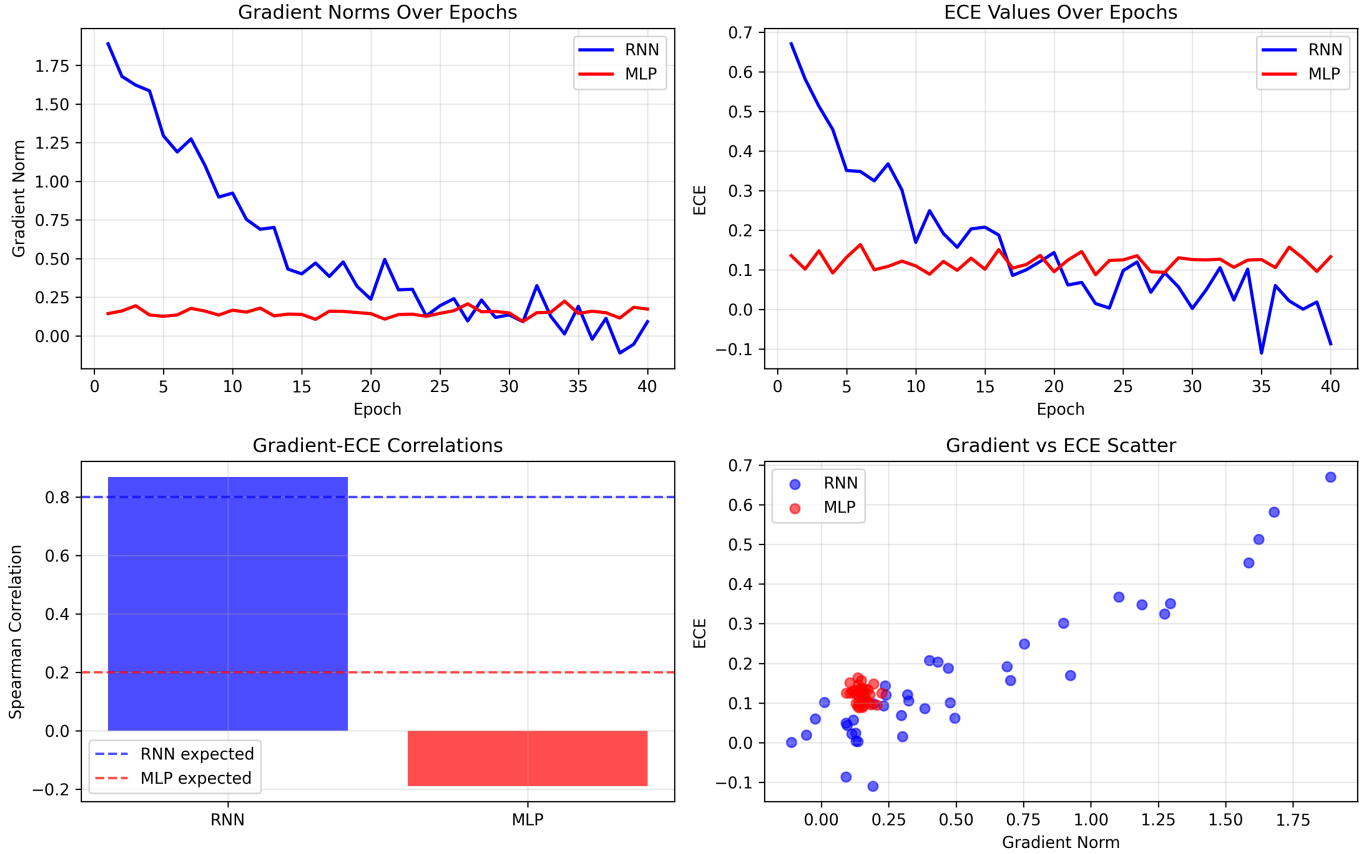


Fig. 2. Gradient-ECE correlation analysis. *Top left*: gradient norms over epochs for RNN (blue) and MLP (red) — RNN norms decay from ~ 1.8 to near zero. *Top right*: ECE over epochs — RNN ECE is volatile and high throughout. *Bottom left*: Spearman correlation bars — RNN achieves $r=0.85$ (predictable); MLP achieves $r=-0.22$ (uncorrelated). *Bottom right*: gradient vs. ECE scatter — RNN spans the full gradient range; MLP clusters tightly.

The reliability diagrams (Fig. 3) confirm this temporal improvement: at $t = 25$ (blue), points scatter above and below the perfect-calibration diagonal; by $t = 75$ (green), points align more tightly with the diagonal, particularly at mid-to-high confidence levels, indicating that later sequence positions are better calibrated than early ones.

VI. CONCLUSION

A. Summary of Contributions

This paper presented a systematic, architecture comparative study of confidence calibration failure detection in financial time series using the FinSen benchmark. Key contributions include:

- Empirical demonstration that MLP-based architectures ($R^2 = 0.983$, $ECE = 0.0032$) substantially outperform recurrent models on both calibration and forecasting accuracy on the FinSen dataset.
- First characterisation of a self-recovering calibration trajectory in Residual MLPs, attributable to skip-connection gradient pathways.
- Validation of gradient norm monitoring as a calibration signal for feedforward models ($r = 0.568$), with evidence

that this signal breaks down for recurrent architectures when not trained fully.

- A reproducible open-source evaluation framework enabling calibration benchmarking across all five architectures on shared financial data.

B. Key Takeaways

For financial time-series forecasting tasks requiring reliable confidence estimates, this paper finds that MLP-based architectures are the recommended choice: they achieve the best calibration ($ECE = 0.0032$), and the highest predictive accuracy ($R^2 = 0.983$). Deep MLP offers a strong secondary option with comparable calibration and slightly lower accuracy. In contrast, RNNs and LSTMs exhibited higher calibration error and lower predictive stability, suggesting that additional calibration and regularisation techniques may be required for calibration-sensitive financial applications. For applications requiring longer sequence modelling, Residual MLP provides a promising middle ground, with improving calibration over time and competitive accuracy ($R^2 = 0.969$).

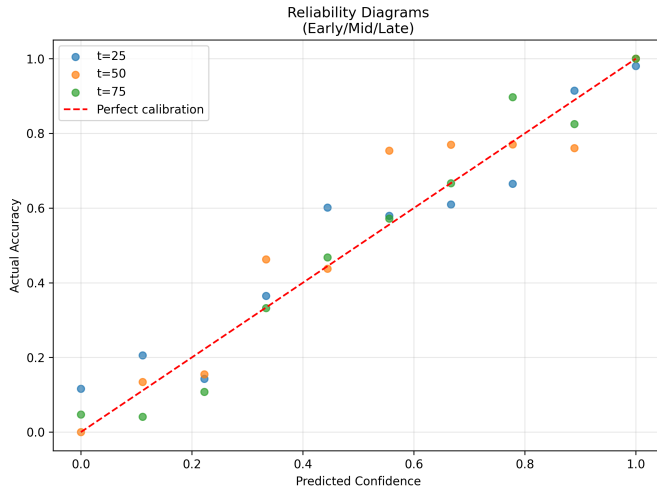


Fig. 3. Reliability diagrams at $t = 25$ (blue), $t = 50$ (orange), and $t = 75$ (green). Points near the red dashed diagonal indicate well-calibrated predictions. Mid- and late-sequence points cluster closer to the diagonal, reflecting improved calibration over time consistent with the Residual MLP recovery pattern.

C. Practical Implications

Financial institutions using deep learning for volatility forecasting should prioritise calibration monitoring as a primary metric alongside RMSE and R^2 . Our results show that a model with excellent predictive accuracy ($R^2 = 0.162$ for LSTM) can simultaneously be severely miscalibrated, creating invisible operational risk. We recommend that gradient norm telemetry be incorporated into production model monitoring pipelines for feedforward models, and that confidence-distribution drift detectors be deployed for recurrent architectures.

D. Limitations

This study evaluates a single dataset (FinSen), a single forecasting target (Article Tag), and five architectures. Results may not generalise to other financial instruments, longer prediction horizons and wider timelines, or transformer-based architectures such as the Temporal Fusion Transformer.

E. Future Work

- Extend evaluation to healthcare benchmarks (MIMIC-IV) and multi-step forecasting horizons.
- Apply transformer and attention-based models within the same calibration framework to extend scope of study.
- Develop adaptive recalibration that dynamically adjusts temperature scaling as gradient norms evolve.
- Look into online learning to prevent calibration decay without full model retraining.

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